

Zero Downtime Linux Updates

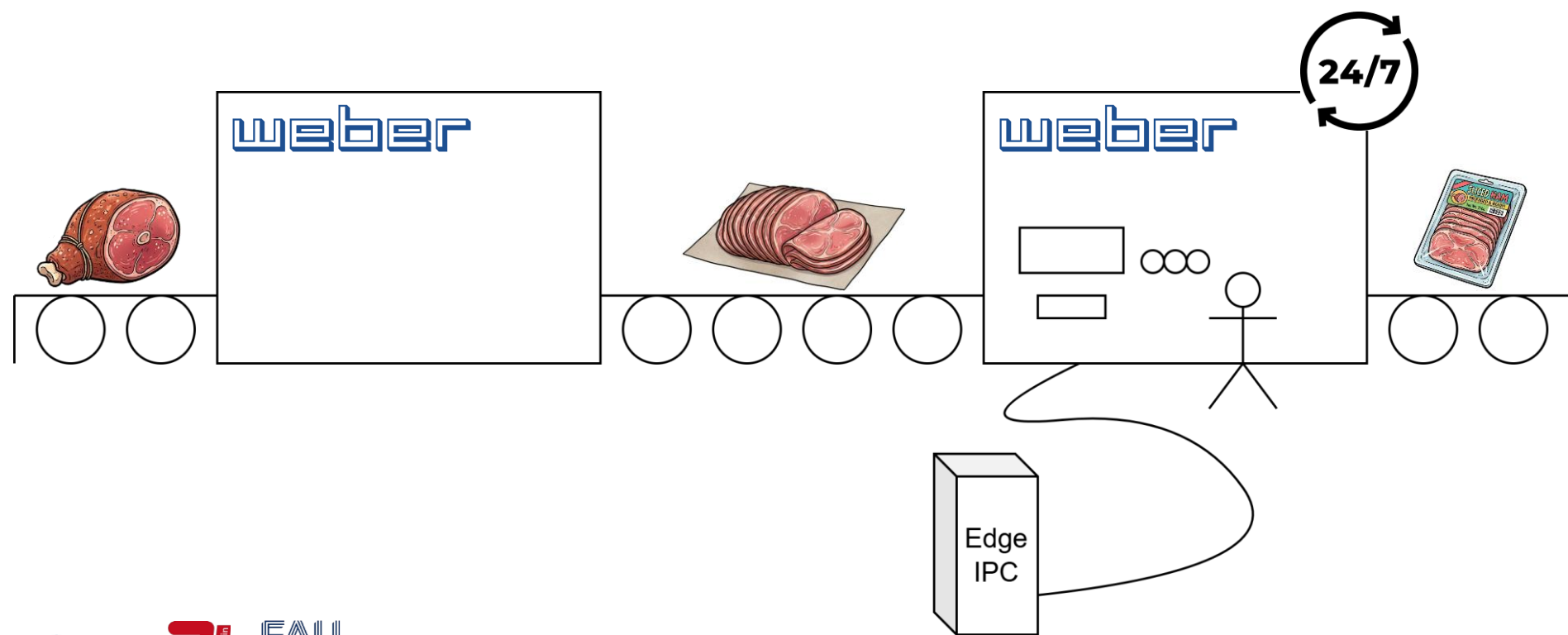
Team



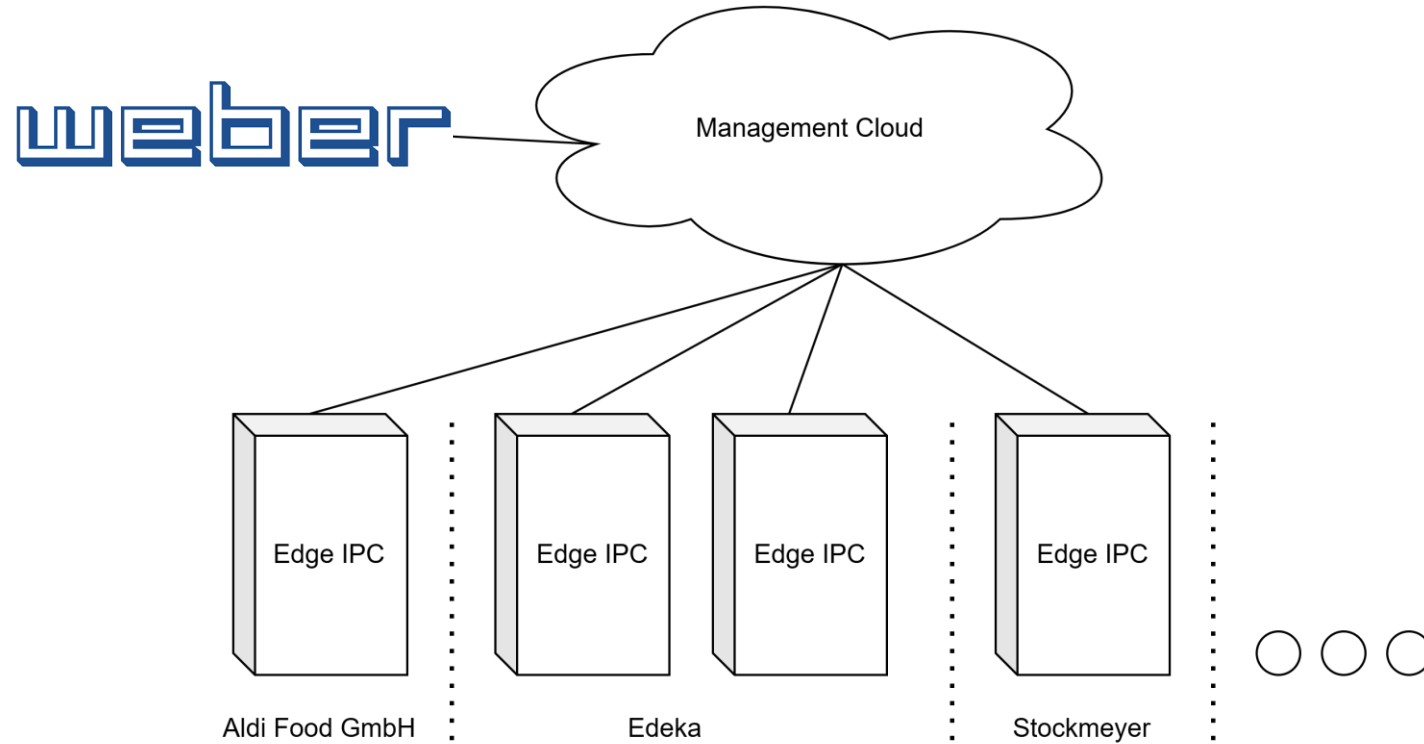
Industry Partner

weber

The Problem

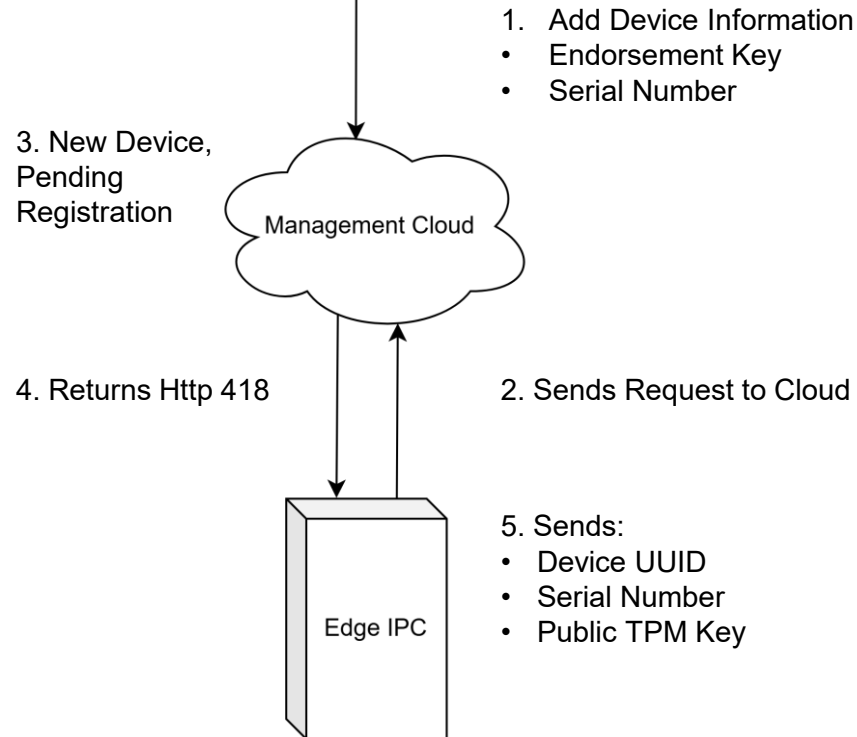


The Solution

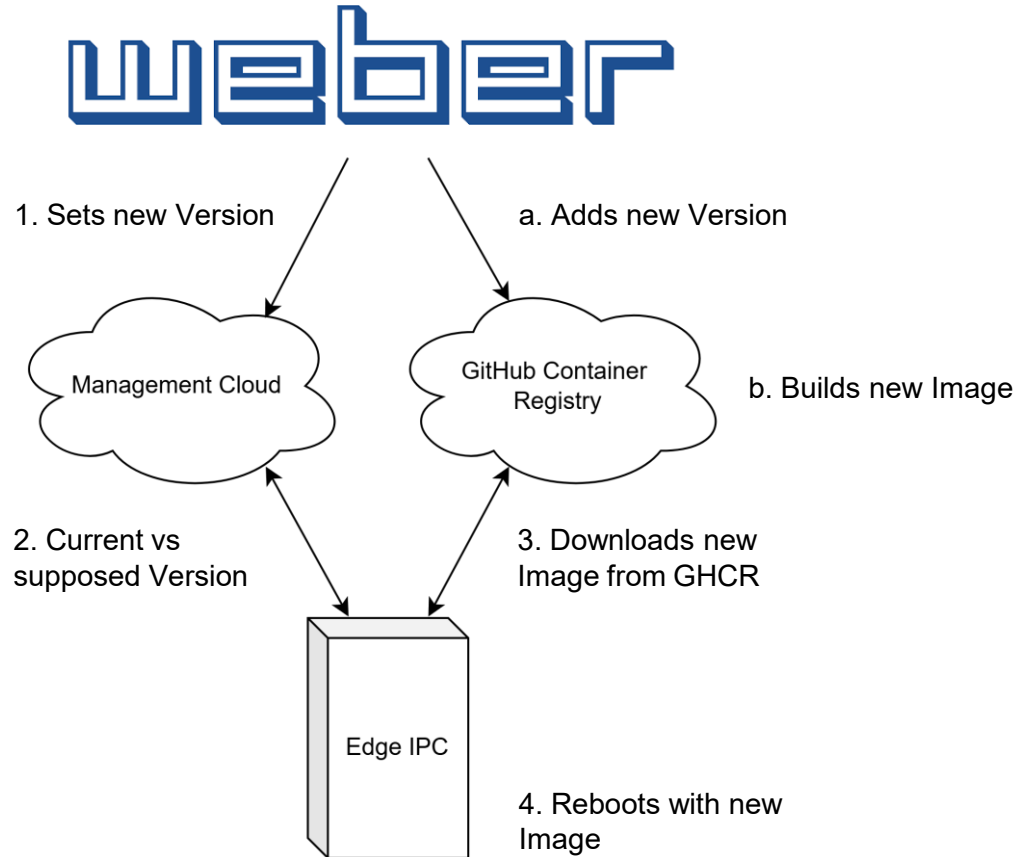


Register Device

weber



OS Swap

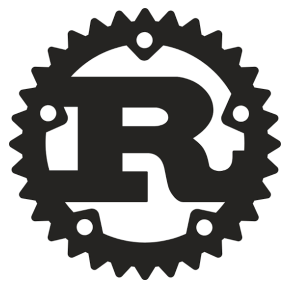


Further Functions

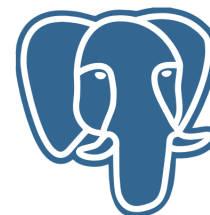
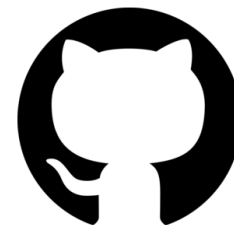
1. Automatic Rollbacks
2. Logging
 - User interaction with DB
 - Device Logs
 - Application Logs
3. Manage Applications
 - Using Podman
 - Assign Application Version to Edge IPC
 - Individual Configs for each Edge IPC (Keys, Names, ...)
4. Create Edge IPC Groups with same Versions



bootc



podman

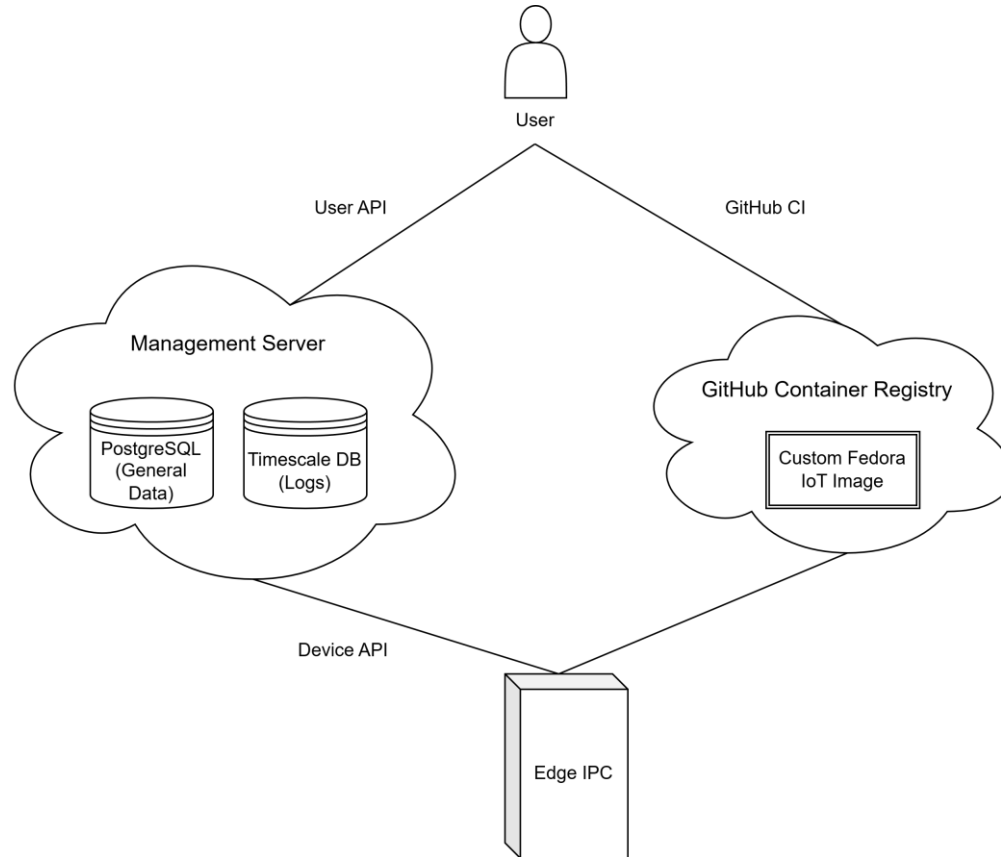


PostgreSQL



TIMESCALEDB

Architecture



Bootc

- Image Based OS
- OCI container image built with a Containerfile like application container
- Install once update with bootc

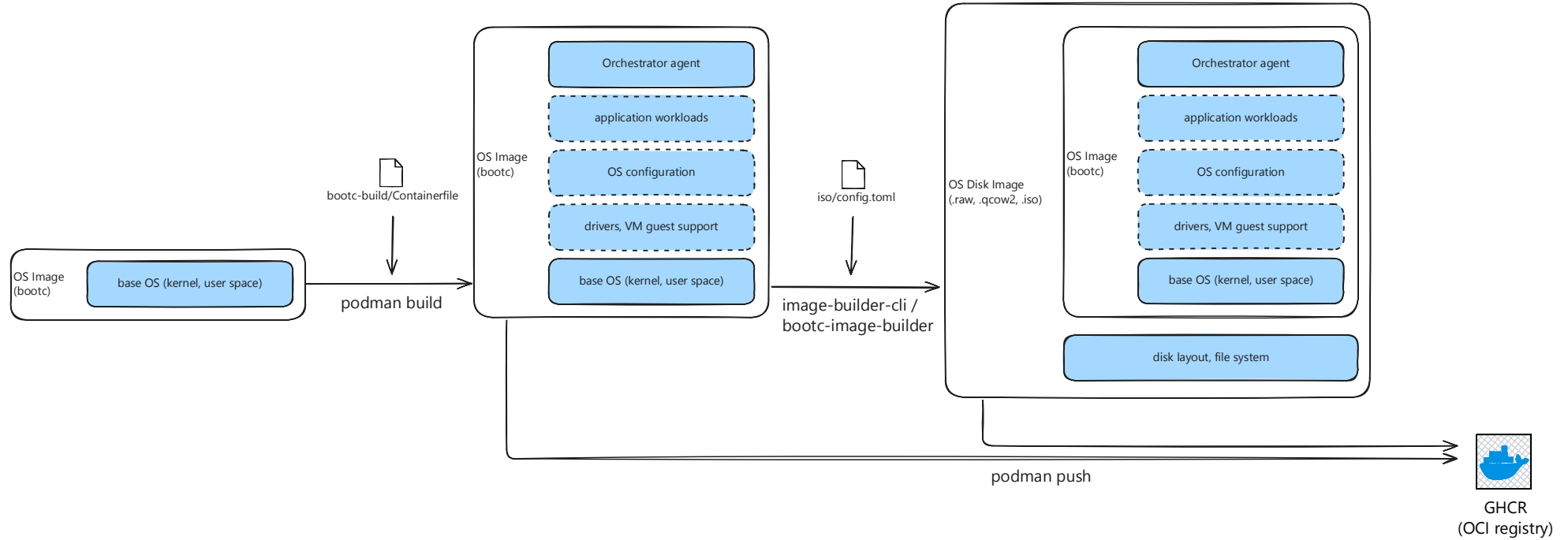
Installation

1. Start from a base bootc image
2. Layer our software
3. Build & push the OS
4. Generate disk

Update

1. bootc switch <image-ref>
2. bootc rollback (or automatic)

Image Build Process



Trusted Platform Module

- Security Microchip
 - Hardware based management of cryptographic keys
 - Used to verify Edge IPC at management cloud
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- TPM creates public key and sends it to management cloud
 - Stored in Database
 - Device sends JWT containing signature
 - Management cloud decode and verifies device with public key